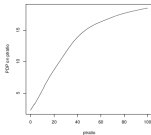
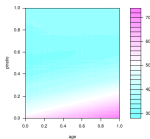
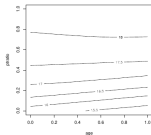
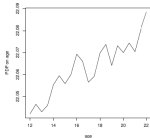
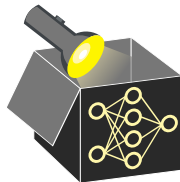


# Interpretable Machine Learning

## Functional Decompositions Introduction



### Learning goals

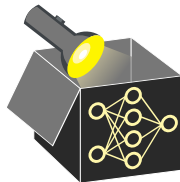
- Basic idea of additive functional decompositions
- Motivation and usefulness of functional decompositions
- Difficulty of obtaining or even defining a functional decomposition
- Several examples

# PRELIMINARIES

## Recap: Interactions

- Interactions between features: Effect of one feature on the prediction output depends on (one or more) other features
- Definition: Features  $x_j$  and  $x_k$  are considered to interact, if

$$\mathbb{E} \left[ \left( \frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_k} \right)^2 \right] > 0$$



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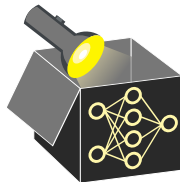
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## Recap: GAMs

- Decomposition into only main effects
- Do not contain any interactions

$$\hat{f}(\mathbf{x}) = \theta_0 + g_1(x_1) + g_2(x_2) + \dots + g_p(x_p)$$



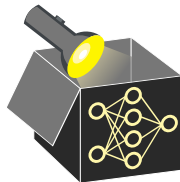
# FIRST EXAMPLE: ADDITIVE DECOMPOSITION

## Example

Consider

$$\hat{f}(x_1, x_2) = 4 - 2x_1 + 0.3e^{x_2} + |x_1|x_2$$

- Idea: Additive decomposition depending on which features used:



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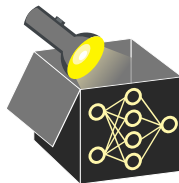
- Idea: Additive decomposition depending on which features used:

$$g_{\emptyset}(x_1, x_2) = 4 \quad \text{Part depending on no features at all (intercept)}$$

$$\left. \begin{aligned} g_1(x_1, x_2) &= 2x_1 \\ g_2(x_1, x_2) &= 0.3e^{x_2} \end{aligned} \right\} \quad \text{Parts depending on a single feature (main effects)}$$

$$g_{1,2}(x_1, x_2) = |x_1|x_2 \quad \text{Part depending on both features (interaction)} \quad (1)$$

→ Single terms with immediate interpretation, full model understanding



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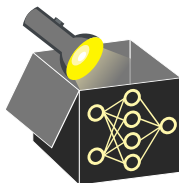
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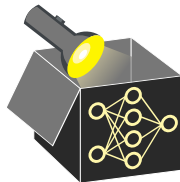
↔ Not possible with effects of single features (e.g. PDPs) or GAM surrogate model (miss interaction part)



# ANOTHER EXAMPLE: ADDITIVE DECOMPOSITION

**Goal** in general: Given a black-box model  $\hat{f} : \mathbb{R}^2 \rightarrow \mathbb{R}$ , find a decomposition

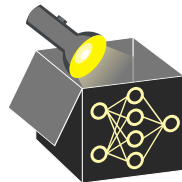
$$\hat{f}(x_1, x_2) = g_{\emptyset} + g_1(x_1) + g_2(x_2) + g_{1,2}(x_1, x_2) \quad (2)$$



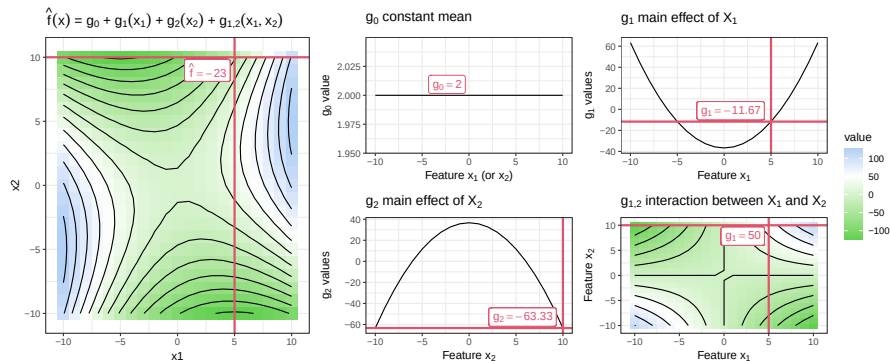
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## Example



→ More details on this example later

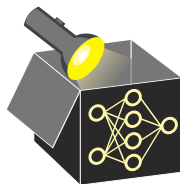


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## Example

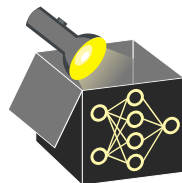
$$\hat{f}(x_1, x_2, x_3) = -2x_1 - 2\sin(x_3) + |x_1|x_2 + 0.5x_1x_2x_3 - \sin(x_2x_3) + 1$$

Again, read additive decomposition from formula:



# ANOTHER EXAMPLE: ADDITIVE DECOMPOSITION

## Example



$$\hat{f}(x_1, x_2, x_3) = -2x_1 - 2\sin(x_3) + |x_1|x_2 + 0.5x_1x_2x_3 - \sin(x_2x_3) + 1$$

Again, read additive decomposition from formula:

|                                                                                                                                                              |                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| $g_{\emptyset}(x_1, x_2, x_3) = 1$                                                                                                                           | constant part, no effects                    |
| $\left. \begin{aligned} g_1(x_1, x_2, x_3) &= -2x_1 \\ g_2(x_1, x_2, x_3) &= 0 \\ g_3(x_1, x_2, x_3) &= -2\sin(x_3) \end{aligned} \right\}$                  | main effects, no interactions                |
| $\left. \begin{aligned} g_{1,2}(x_1, x_2, x_3) &=  x_1 x_2 \\ g_{1,3}(x_1, x_2, x_3) &= 0 \\ g_{2,3}(x_1, x_2, x_3) &= -\sin(x_2x_3) \end{aligned} \right\}$ | 2-way interactions (depending on 2 features) |
| $g_{1,2,3}(x_1, x_2, x_3) = 0.5x_1x_2x_3$                                                                                                                    | 3-way interactions                           |

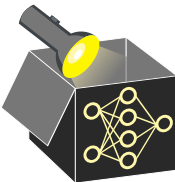
(3)

$\Rightarrow$  8 components in total, but some empty  $\rightsquigarrow$  Certain interactions not present

# GENERAL FORM OF FUNCTIONAL DECOMPOSITION

► "Li and Rabitz" 2011

► "Chastaing et al." 2012



## Definition

*Functional decomposition:* Additive decomposition of a function  $\hat{f} : \mathbb{R}^p \mapsto \mathbb{R}$  into a sum of components of different dimensions w.r.t. inputs  $x_1, \dots, x_p$ :

$$\begin{aligned}\hat{f}(\mathbf{x}) &= \sum_{S \subseteq \{1, \dots, p\}} g_S(\mathbf{x}_S) \\ &= g_{\emptyset} + g_1(x_1) + g_2(x_2) + \dots + g_p(x_p) + \\ &\quad g_{1,2}(x_1, x_2) + \dots + g_{p-1,p}(x_{p-1}, x_p) + \dots + \\ &\quad g_{1,2,3}(x_1, x_2, x_3) + \dots + g_{1,2,3,4}(x_1, x_2, x_3, x_4) + \dots + g_{1,\dots,p}(x_1, \dots, x_p)\end{aligned}$$

$\rightsquigarrow$  one component for every possible combination  $S$  of indices, allowed to formally depend on these features / be a function of these features

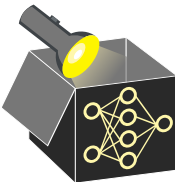
## Problems:

- How to find / compute such a decomposition for any black-box models  $\hat{f}$ ?
- ... such that the decomposition is useful / has nice properties (w.r.t. the model / w.r.t. the data)?

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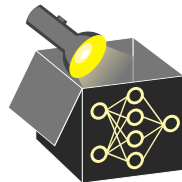
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Sort terms according to degree of interaction:

- $g_{\emptyset} \hat{=}$  Constant mean (intercept)
- $g_j \hat{=}$  first-order or main effect of  $j$ -th feature alone on  $\hat{f}(\mathbf{x})$
- $g_{j,k} \hat{=}$  second-order interaction effect of features  $j$  and  $k$  w.r.t.  $\hat{f}(\mathbf{x})$
- $g_S(\mathbf{x}_S) \hat{=}$   $|S|$ -order effect, depends **only** on features in  $S$

# PROPERTIES OF THE DEFINITION

- **Interpretability:** Extremely powerful decomposition, reveals complete interaction structure

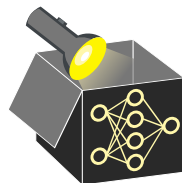


# PROPERTIES OF THE DEFINITION

- **Interpretability:** Extremely powerful decomposition, reveals complete interaction structure
- Compare to GAM: Same decomposition, but without interactions  
⇒ Any GAM already comes with its decomposition

$$\hat{f}(\mathbf{x}) = g_{\emptyset} + g_1(x_1) + g_2(x_2) + \dots + g_p(x_p)$$

- Same for LMs: Decomposition explicitly modeled

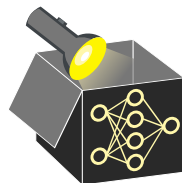


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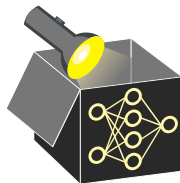


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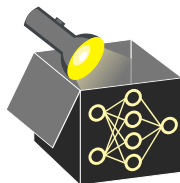


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- **Problem 2:** Definition not complete: Decomposition not unique, many trivial decompositions not useful
  - More requirements or constraints needed to ensure decomposition is meaningful
    - Even worse once features are dependent or correlated (see later)

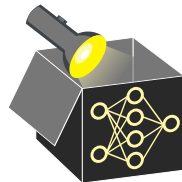


# PROBLEM 2: DEFINITION NOT ENOUGH

## Example

Again consider

$$\hat{f}(x_1, x_2, x_3) = -2x_1 - 2\sin(x_3) + |x_1|x_2 + 0.5x_1x_2x_3 - \sin(x_2x_3) + 1$$



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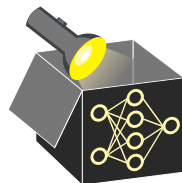
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Two possible decompositions (valid according to definition):

$$g_{1,\dots,p}(x_1, \dots, x_p) := \hat{f}(\mathbf{x}) \text{ and for all other terms } g_S(\mathbf{x}_S) := 0,$$

or:

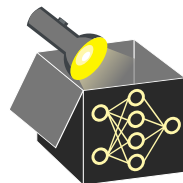
$$\begin{aligned} g_{\emptyset} &= 1; \quad g_1(x_1) = x_1; \quad g_2(x_2) = 2x_2; \quad g_3(x_3) = 3x_3; \\ g_{1,2}(x_1, x_2) &= \frac{1}{2}x_1x_2; \quad g_{1,3}(x_1, x_3) = \frac{1}{3}x_1x_3; \quad g_{2,3}(x_2, x_3) = \frac{2}{3}x_2x_3; \\ \text{and } g_{1,2,3}(x_1, x_2, x_3) &= \hat{f}(x_1, x_2, x_3) - \sum_{S \subsetneq \{1,2,3\}} g_S(\mathbf{x}_S) \end{aligned}$$



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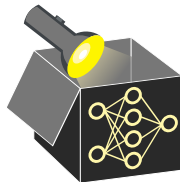
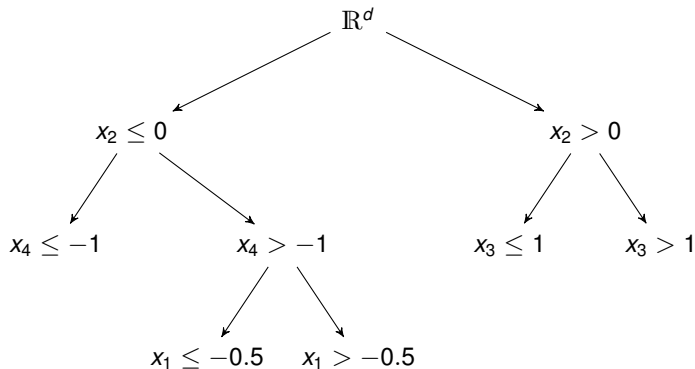
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$\implies$  Definition of decomposition not unique

# EXAMPLE: DECISION TREES

Define *interaction type  $t$*  of a node: subset of features used to build this node.

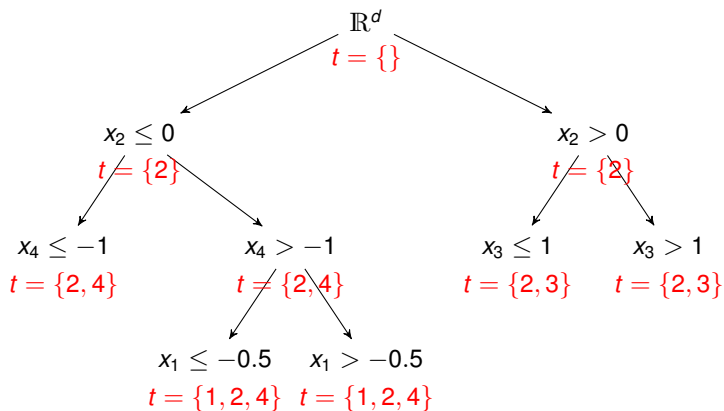
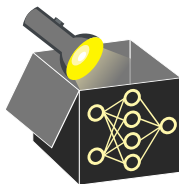
**Example:**



# EXAMPLE: DECISION TREES

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**Example:**



⇒ Degree of interaction in each node is  $|t|$ .

# DECOMPOSITION FOR DECISION TREES

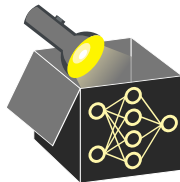
Here: Decomposition via indicator functions

$$\hat{f}(\mathbf{x}) = g_{\emptyset} + g_{2,4}(x_2, x_4) + g_{2,3}(x_2, x_3) + g_{1,2,4}(x_1, x_2, x_4)$$

⇒ Decomposition has no main effect, but model certainly contains an effect of  
e.g.  $x_2$

⇒ Lower-order effects “hidden” inside higher-order terms

↪ reading from decision tree not enough, “bad decomposition”



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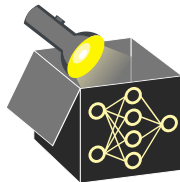
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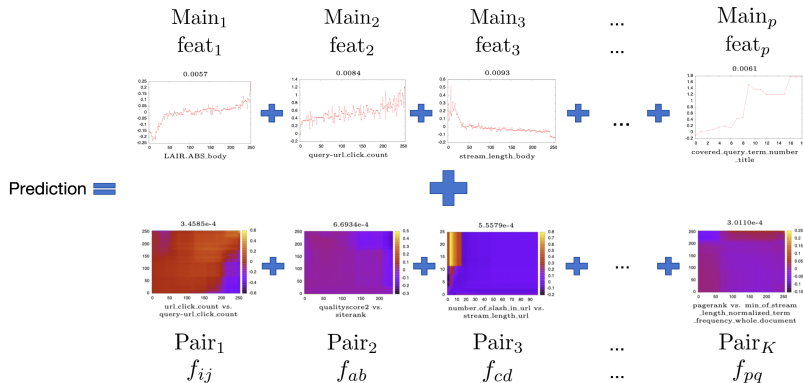
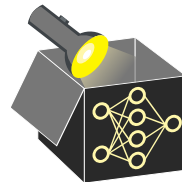
**Note:** ▶ “Yang” 2024 propose a (quite complicated) solution for this case





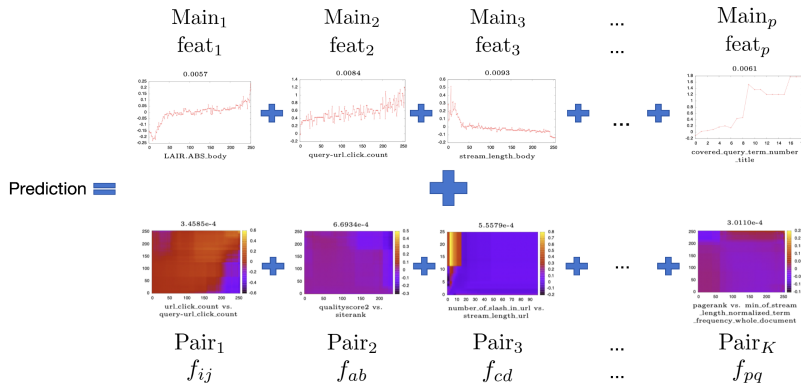
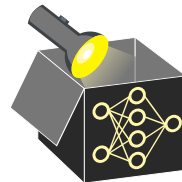
# EXAMPLE: EBM

- See before: **GAMs** have functional decomposition by definition
- **EBMs**: Sum of the final one- and two-dimensional components



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- See before: **GAMs** have functional decomposition by definition
- **EBMs**: Sum of the final one- and two-dimensional components



- In general: Model with functional decomposition up to max. order 2 is always “inherently interpretable”
- **Reason:** Visualization of all components